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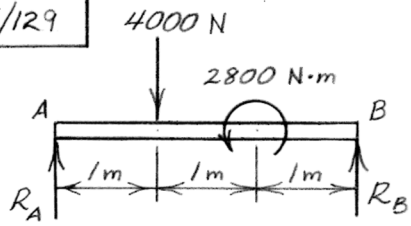


$\circlearrowleft + \Sigma M_A = 0: \; -4000(1) + 2800 + R_B(3) = 0$

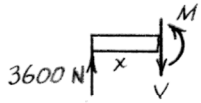
$R_B = 400 \text{ N}$

$+\uparrow \Sigma F_y = 0: \; R_A - 4000 + 400 = 0$

$R_A = 3600 \text{ N}$

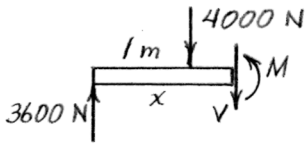
or 3.6 kN

$0 < x < 1 \text{ m}:$



$V = 3600 \text{ N}$

$M = 3600x$

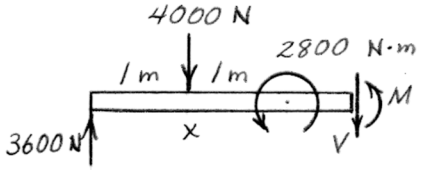


$1 \text{ m} < x < 2 \text{ m}:$

$+\uparrow \Sigma F = 0: \; 3600 - 4000 - V = 0, \; V = -400 \text{ N}$

$\circlearrowleft + \Sigma M = 0: \; -3600x + 4000(x-1) + M = 0$

$M = 4000 - 400x$

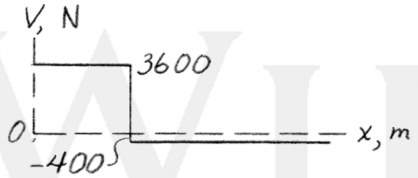


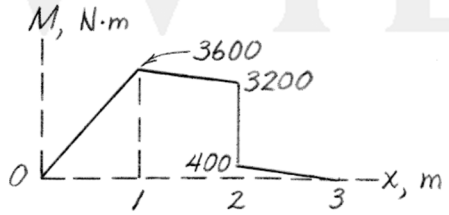
$2 \text{ m} < x < 3 \text{ m}:$

$\Sigma F = 0: \; 3600 - 4000 - V = 0$

$V = -400 \text{ N}$

$\Sigma M = 0: \; -3600x + 4000(x-1) + 2800 + M = 0, \quad M = 1200 - 400x$





Values at middle:

$V = -400 \text{ N}$

$M = 3400 \text{ N·m}$